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3/30/2021

CSE 13S Spring 2021
Assignment 1: Left, Right and Center
Design Document

In the game *Left, Right, and Center*, there are 2 to 14 players that each have $3 and three dice with 6 faces each. 3 of the faces are a •, 1 is an L (left), 1 is an R (right), and 1 is a C (center). Player 1 rolls the dice a number of times equal to the number of dollars he or she has and passes if he or she has no money.
For each die:

- If the player rolls L, he gives $1 to the player on his left.
- If the player rolls R, he gives $1 to the player on his right.
- If the player rolls C, he puts the $1 in the pot in the center.
- If the player rolls •, he ignores it.

Then move to the next player to the right. Repeat until only one player has any money. That remaining player wins the pot in the center.

I will implement this game by using the random() function to simulate rolling a dice. The range will be from 0-5 to represent the six faces of the dice.

The inputs to our program are:

1. The random seed
2. The number of players

When using the same random seed and the same number of players, the game and answer will always be the same.

TOP LEVEL:

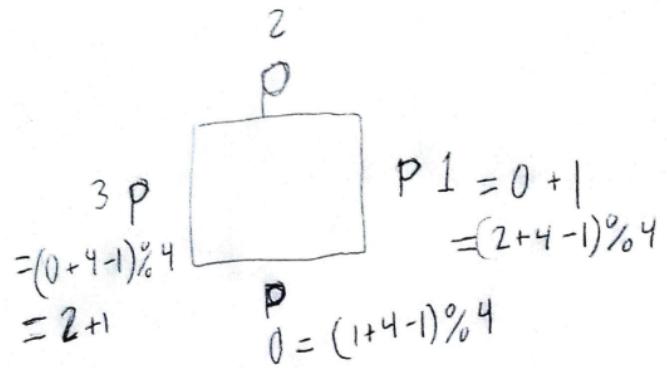
The top level design of the code is given by the following pseudocode:

Main:

```
    Read the random seed
    Read the number of players
    If the number of players k is not > 1 and < or == 14:
        Print("Number of players must be from 1 to 14.")
        End program

    For each player:
        For each dollar that the player has:
            Player rolls a die
```

If player rolls L:
    Give $1 to player on left
If player rolls R:
    Give $1 to player on right
If player rolls C:
    Put $1 in the pot in the center
If only one player has any money:
    That player wins
    End program



P0 roll x3
P1 roll x3
P2 roll x3
P3 roll x3

num rolls = num $

at start: num alive = num players

if player x has $0, num alive -=1

if player with $0 is given $1, num alive +=1

if num alive = 1, end game